A320 FAL TWIN · DECISION TRACE

FINAL ASSEMBLY LINE S50 → FLIGHT LINE · INTERACTIVE CONCEPT

One-pager · October 2026 · Miguel Figueroa
github.io · a320-fuselage-agent-control/fal-twin

SYNTHETIC DATA

GOAL

WHY THIS TWIN EXISTS

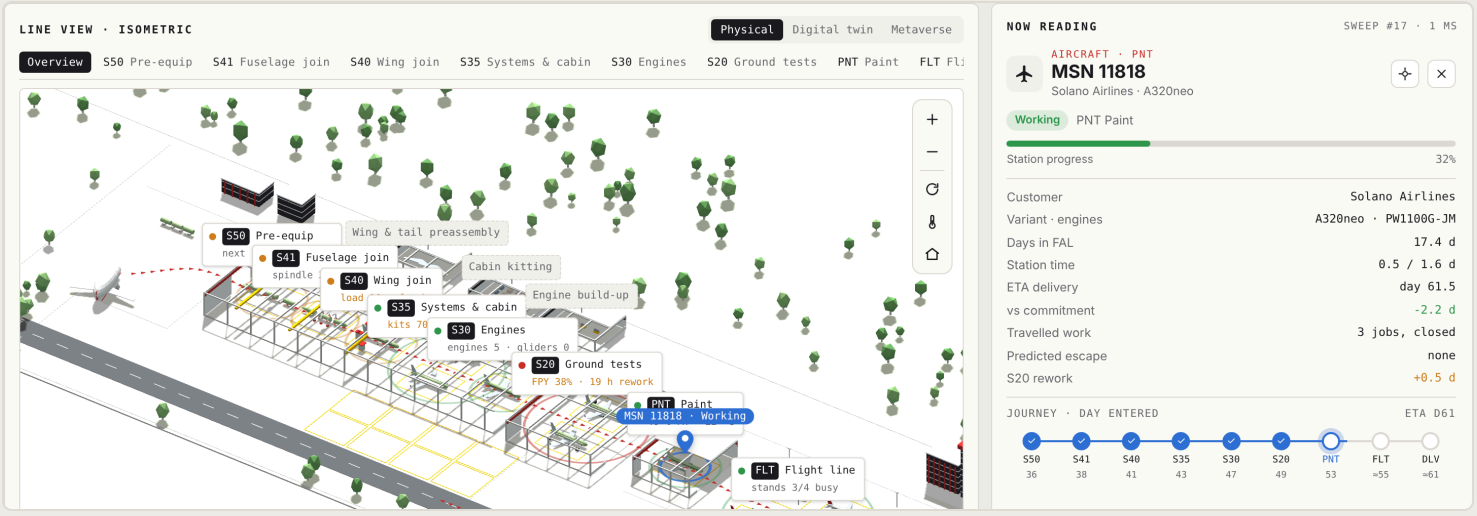
Make A320 final assembly **more efficient**: reach takt at **10.5** aircraft per month, deliver every aircraft within its **committed lead time**, and remove rework and travelled work at the source.

It shows where time is lost and tests each fix in simulation first, so every decision comes with a measured effect on rate, lead time and quality.

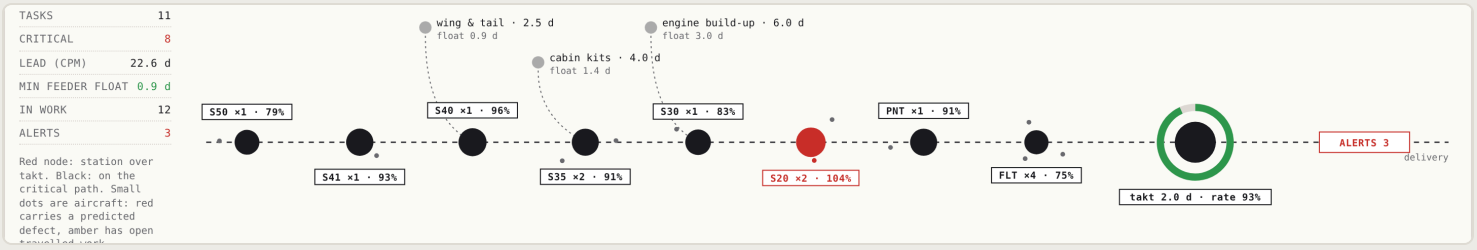
KEY RESULTS · NOW VS TARGET

TYPICAL RUN · TARGETS ILLUSTRATIVE

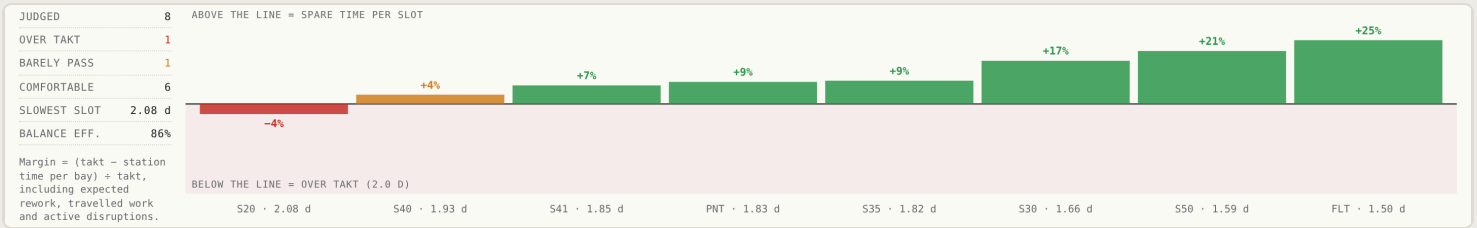
KPI	NOW	TARGET	LEVER IN THE TWIN
Delivery rate	9.8 /mo	≥ 10.5	third S20 shift, floating team
Lead time	23.6 d	≤ 26.7 d	critical path, line rebalancing
On-time delivery	100 %	≥ 90 %	late-MSN alerts, rush-order planner
First-pass yield	38 %	≥ 80 %	edge-AI hole inspection, kit gate
Rework	19 h/ac	≤ 8 h	edge-AI inspection at S41
Gliders	0	0	engine-supply alert, expedite
OEE line	33 %	≥ 60 %	quality levers lift Q, balance lifts A
Power peak · 24 h	2,630 kW	≤ 2,650	stagger cure ovens, battery shaving



Line view (isometric 3D, physical · digital twin · metaverse · heat) beside "Now reading": the selected aircraft with status, ETA vs commitment and its station-by-station journey.



Relation graph: feeders → stations → takt → alerts. Node size = load vs takt; red = over takt, black = critical path; small dots are aircraft.



Takt margin field: each station's distance to the takt line, sorted worst to best; below the line = over takt.

HOW A DECISION FLOWS

- **SENSE** · station progress, robot spindles, booth climate, supplier ETAs and power update every bay and aircraft.
- **DETECT** · rules flag bottlenecks, late MSNs, engine gaps and sensor trends, e.g. robot R-3 reaching its vibration limit.
- **DECIDE** · the assistant ranks options with simulated gain and cost; the rush-order planner proposes slots and buffers.
- **PROVE** · the impact report re-runs plan vs change for 200 d × 3 seeds and logs each decision for the shift handover.

BUILD & RUN

STACK	One self-contained HTML page: three.js scene, discrete-event simulation and CPM in the browser. No backend.
ASSISTANT	Rule-based on the twin's model, not a language model; nothing changes until you apply.
WALL DISPLAY	Keeps the screen awake (wake lock + video fallback), fullscreen with F.
LIVE	mfmadinaveitia1-del.github.io/a320-fuselage-agent-control/fal-twin/

All layouts, cycle times, rates, defect rates, costs, customers and power figures are synthetic, invented for this concept. They do not represent any manufacturer's data. KPI values are from one typical simulation run.